

Mapping of sea ice using the NIMBUS 6 satellite SCAMS microwave radiometer

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Abstract

We analyze SCAMS microwave radiometer observations from March 1976 to evaluate sea ice concentration (SIC) retrievals using both one-channel and two-channel algorithms. A radiative transfer model (RTM) is applied to correct brightness temperatures, enabling assessment of atmospheric effects on the retrievals. Sensitivity analysis shows that cloud liquid water exerts by far the strongest influence on the 22 and 31 GHz channels, followed by water vapor, wind speed, and incidence angle, while sea surface temperature and air temperature contribute only weakly in polar conditions. Applying the correction generally increases SIC estimates, reduces brightness temperature variability, and improves consistency between the one- and two-channel algorithms. The resulting SIC fields and ice-type classifications reveal physically plausible patterns, with remaining discrepancies likely linked to sampling limitations and model uncertainties.

1 Introduction and Background

The objective of this report is to retrieve sea ice concentration and ice type data for March 1976 by applying one- and two-channel sea algorithms to data from the SCAMS microwave radiometer aboard the Nimbus-6 satellite mission. A radiative transfer model together with ERA5-reanalysis data are used to apply an atmospheric correction to the raw brightness temperature measurements.

1.1 NIMBUS-6 Mission Overview

Nimbus-6 was launched by NASA on 12 June 1975 and operated as part of the second generation of the Nimbus meteorological research satellites. The spacecraft flew in a near sun-synchronous polar orbit at an altitude of approximately 1101 km. Sun-synchronous means, that the satellite passes the same local spot at the same time everyday (European Space Agency [2020](#)). According to the mission summary provided by the ESA Earth Observation Handbook, the satellite had an orbital inclination of 99.9° and an orbital period

of 107.4 min, ensuring global coverage through a stable repeat cycle (ESA Earth Observation Handbook [2025](#)). Nimbus-6 remained operational until 29 March 1983, substantially exceeding its nominal lifespan.

The Nimbus program (1964-1993) was fundamental in establishing early satellite-based atmospheric and climate observations. Nimbus-6 carried different radiometric and atmospheric sounding instruments, including the Scanning Microwave Spectrometer (SCAMS), the Electrically Scanning Microwave Radiometer (ESMR), the Temperature-Humidity Infrared Radiometer (THIR), and the Satellite Infrared Spectrometer (SIRS). These instruments provided data on atmospheric temperature and humidity profiles, cloud structure, microwave surface emissivity, and sea-ice concentration.

Of particular relevance to this study is the SCAMS instrument, which provided brightness temperature measurements at 22 GHz and 31 GHz. These channels are dominated by surface emission over polar oceans and sea ice, making Nimbus-6 a valuable platform for historical sea-ice mapping and for developing atmospheric correction and retrieval algorithms used in this work (Kolbe, Tonboe, and Stroeve [2025](#)).

1.2 SCAMS Instrument Description

NIMBUS had 9 different experiments on board. One of them was the Scanning Microwave Spectrometer (SCAMS). SCAMS was a multi frequency radiometer scanning at five different frequency channels. The original purpose of SCAMS was to globally map the tropospheric temperature profiles and also map the liquid water and water vapor in the atmosphere over the worlds oceans. The big difference from NEMS which was the predecessor microwave spectrometer that was mounted on NIMBUS 5 is the different scan angles of SCAMS. Where NEMS was a one-channel radiometer that could only scan at nadir, SCAMS has 13 scan angles. The amount of angles and the resulting broad swath made it possible for SCAMS to nearly cover the entire earth in 12 hours. (LANDSAT/Nimbus Project ([1975](#)))

1.3 ERA-5 Reanalysis Data

ERA-5 is a global atmospheric reanalysis product created by the ECMWF. It provides a physically consistent reconstruction of the Earth system from 1959 to the present day (Hersbach [2020](#)). A wide range of in situ and satellite observations are assimilated into a numerical weather prediction model to generate hourly fields with a grid resolution of approximately 31×31 km ([bell2021hadisst2](#)). In this project, ERA-5 provides three key atmospheric correction variables for the SCAMS brightness temperatures: total column water vapor, sea ice concentration, and 2-meter air temperature. Total column water vapor is particularly important because it is the main source of atmospheric noise affecting the 22 GHz and 31 GHz SCAMS channels. Variations in this area are therefore later in this report used in conjunction with a radiation transfer model to minimize atmospheric disturbances in brightness temperatures. (Kolbe, Tonboe, and Stroeve [2025](#)). Sea ice concentration from ERA-5, based on HadISST2.0 prior to 1979 and informed by digitized historical ice charts, serves as an external reference for distinguishing between open

water and ice surfaces during tie-point selection (bell2021hadisst2). The 2-metre air temperature is also used to identify inconsistencies in the collocated SCAMS observations and to support quality control procedures (Kolbe, Tonboe, and Stroeve 2025). ERA-5's temporal and spatial resolution make it particularly suitable for collocating atmospheric information with the SCAMS swath data. In the SCAMS–ERA-5 processing chain, each satellite footprint is matched to the nearest ERA-5 grid cell in space and time (Kolbe, Tonboe, and Stroeve 2025). This ensures that the atmospheric state relevant to each observation is accurately represented. This collocation enables brightness temperatures to be corrected for water-vapor-induced variability and ensures robust separation between ice and open-water signatures when deriving sea ice concentration. As described in the processing framework, ERA-5 therefore plays an essential role in reducing regional noise, defining tie-points (specific point on the ground which can be clearly identified in two or more images), and improving the physical consistency of the derived sea ice concentration (Kolbe, Tonboe, and Stroeve 2025).

2 Methodology

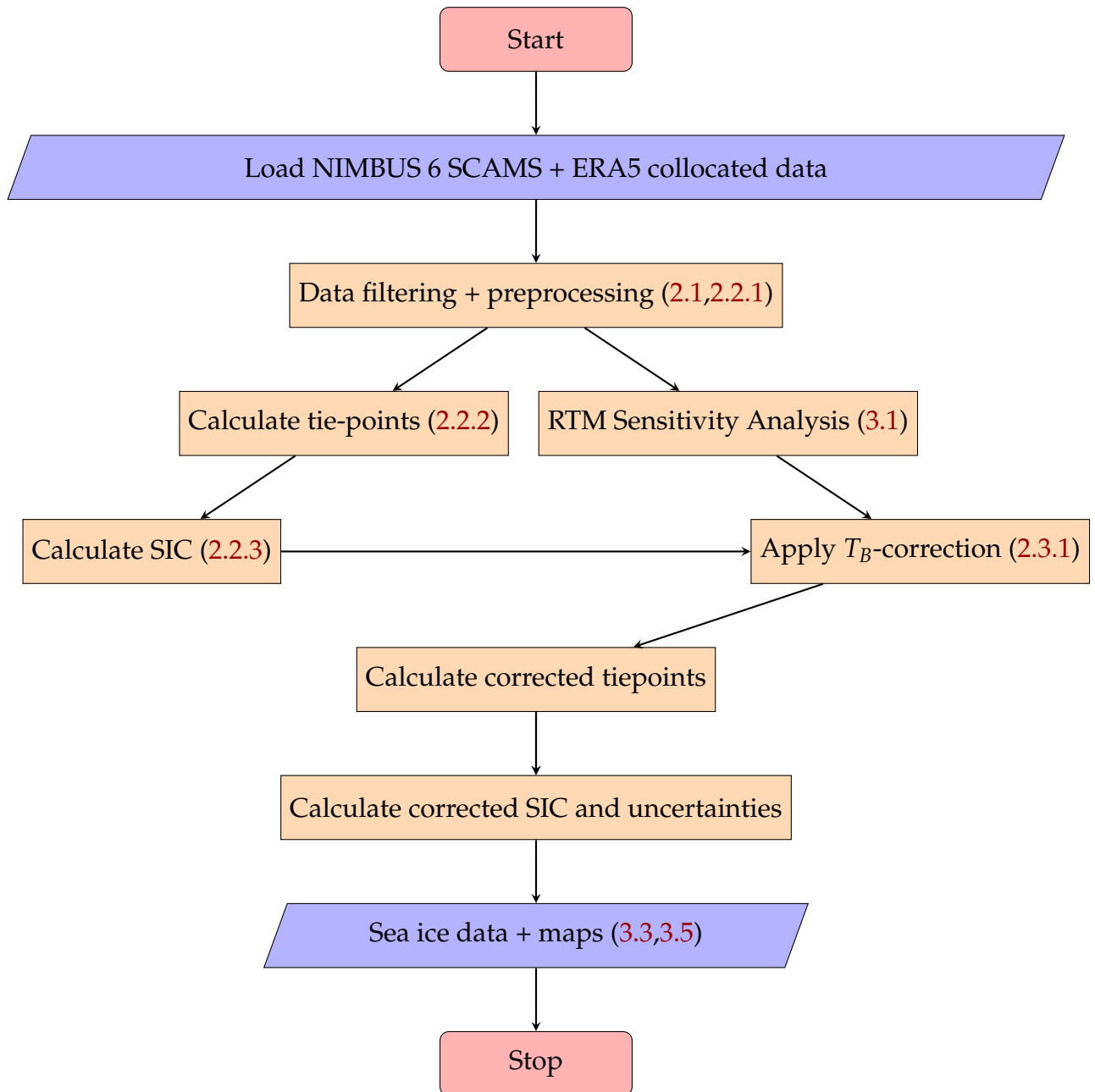
The SCAMS data is collocated with the corresponding atmospheric variables from the ERA5 reanalysis. The processing pipeline begins by loading the SCAMS files, and filter out the files with erroneous timestamps and errors in the brightness temperature (T_B).

After filtering, the remaining data are used to calculate daily tie points for open water and sea ice for the one-channel algorithm, and daily tie-points for first year ice and multi year ice for the two channel algorithm. These tie-points are then used to calculate the uncorrected sea ice concentration (SIC) using both the one-channel and the two-channel algorithm.

The filtered data, together with the collocated atmospheric data from ERA5 also serve as input in the radiative transfer model (RTM) sensitivity analysis, where the atmospheric parameters are tested individually to better understand which parameters the transfer model is most sensitive to.

In the atmospheric correction step, the uncorrected one-channel SIC and ERA5 variables are used as input to the RTM. The RTM is then run twice. Once using mean ERA5 values to obtain a reference T_B , and once using the specific ERA5 values for each pixel. The difference between the two is used as an atmospheric correction that is added to the measured T_B .

After the T_B -correction is applied it is necessary to recalculate the tie-points and the SIC with the corrected tie-points. This results in a final one-channel and two-channel SIC that is less affected by atmospheric noise. Finally, the sea ice data are visualized in several figures and maps for the desired analysis.



2.1 Preliminary Data Filtering

The SCAMS data is ordered in netCDF files and it was necessary to go through this data and filter out errors in the data. The first step was to assess the time stamps of the data. Since the orbit time of the NIMBUS 6 satellite is about 107 minutes, a single file should only cover the same amount of time. 107 minutes correspond to a maximum of three different hour-stamps. Checks were implemented and the bad files removed from the dataset. This resulted in 317 good files and 54 discarded files. The next filter was

implemented to remove files containing errors in one of the two brightness temperature channels. For each beam position it was necessary to check if more than 25% of the data-points have the exact same brightness temperature values. 15 files were removed in this filtering step. We concluded that no additional correction for overlapping record entries was necessary.

2.2 Sea Ice Concentration Retrieval Algorithms

Before discussing methodologies for retrieval of sea ice concentration from radiometric data, it is necessary to first define some terms. Sea ice concentration is defined as the fraction of a specific ocean area covered by ice of any type and any thickness. Sea ice concentration algorithms make use of the linear mixture assumption, meaning that each satellite footprint is assumed to be a linear mixture of ice types and water. Sea ice concentration can thus be estimated by linearly mapping the measured brightness temperatures onto a scale between water and ice. This is done by first determining "reference" brightness temperatures called *tiepoints* which are characteristic of water or various types of ice. This type of indirect measurement is possible because water and ice exhibit different emissivities due to their differing electromagnetic properties: for example, water exhibits a large difference in emissivity as a function of polarization (H or V), whereas this difference (AKA polarization ratio) is relatively small in ice. (Carrieres et al. 2017)

One channel algorithms use brightness temperature measurements at one specific frequency and polarization to produce a sea ice concentration measurement of generic ice, whereas multi-channel algorithms utilize either a frequency ratio or a polarization ratio between two different brightness temperature measurements to produce a sea ice concentration that further specifies the ice type (Carrieres et al. 2017). For the northern hemisphere, tie points are generally computed for two ice types: first year ice (FYI), and multi-year ice (MYI). For the southern hemisphere, the corresponding tie points are instead recognized as two snow and ice surfaces with different radiometric signatures, typically denoted A (FYI) and B (MYI) (Kolbe, Tonboe, and Stroeve 2025).

In this paper, we calculate sea ice concentration using both a single channel algorithm and the two channel algorithm specified in (Comiso et al. (1997)). The following section will detail the preprocessing steps. The code for these steps can be found on the project [github](#) in the file `seaicemapping.ipynb`.

2.2.1 Preprocessing

As described previously, the original SCAMS data contain approximately one orbit per .nc file. Before tiepoints can be calculated, it is therefore necessary pre-process the data such that we obtain a new directory containing one file per day for our sample period encompassing March 1976. This processing step is illustrated in figure 1.

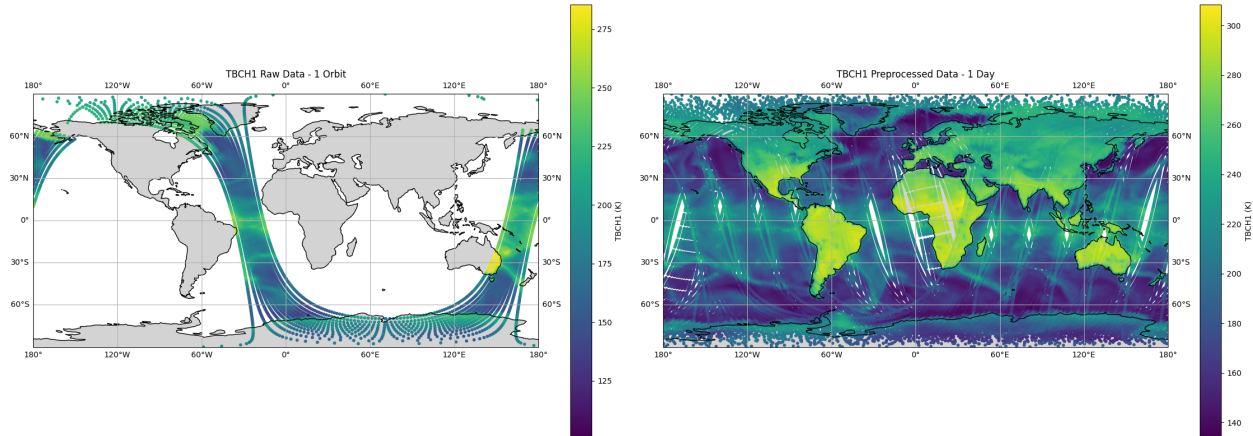


Figure 1: Left: original input file containing one orbit, showing the 13 scan angles composing the swath. Right: preprocessed input file consisting of all orbit data for one day, showing global coverage.

2.2.2 Tiepoint Calculation

Once the preprocessed input of one file per day in the sample month is obtained, the next step was to calculate the tiepoints used for both the single channel and two channel algorithms. The purpose of doing this is to calibrate the algorithms to the dataset, removing potential bias from sources like an incompletely calibrated radiometer or seasonal variations that change over time. The tiepoints were computed as daily averages for each of the 13 scan angles, such that each day of data contains 13 tie points for each water/ice classification. Furthermore, the tiepoints were computed separately for the northern and southern hemispheres.

Consequently, separate masking conditions were needed to compute each set of tiepoints. The ERA5 land-sea mask proved insufficient in fully excluding points on land, likely because it has coarse resolution, but this was resolved by implementing an additional masking condition for elevation less than 0.05 m. Otherwise, the masking conditions were identical to those described in (Kolbe, Tonboe, and Stroeve (2025)), and are reproduced in table 4. For the two channel algorithm, the gradient is used to differentiate between FYI and MYI and is defined as follows:

$$GR = \frac{T_{B,31} - T_{B,22}}{T_{B,31} + T_{B,22}} \quad (1)$$

Identically to (Kolbe, Tonboe, and Stroeve (2025)), we use a gradient threshold of -0.015, where footprints with gradients higher than this threshold were classified as first year ice and footprints with gradients lower than this threshold were classified as multi year ice.

The final filtering step for both the FYI and MYI masks describes the outermost beam exclusion for latitudes under the poles, which is done in order to reduce noise introduced by the larger incidence angle of the outermost beams. An example result of this masking used for validation is shown in figure 2. Once the masks were defined, the water, ice (1

channel), FYI (2 channel), and MYI (2 channel) tiepoints for the arctic and antarctic regions for each set of scan angles for each day in the sample period could be determined.

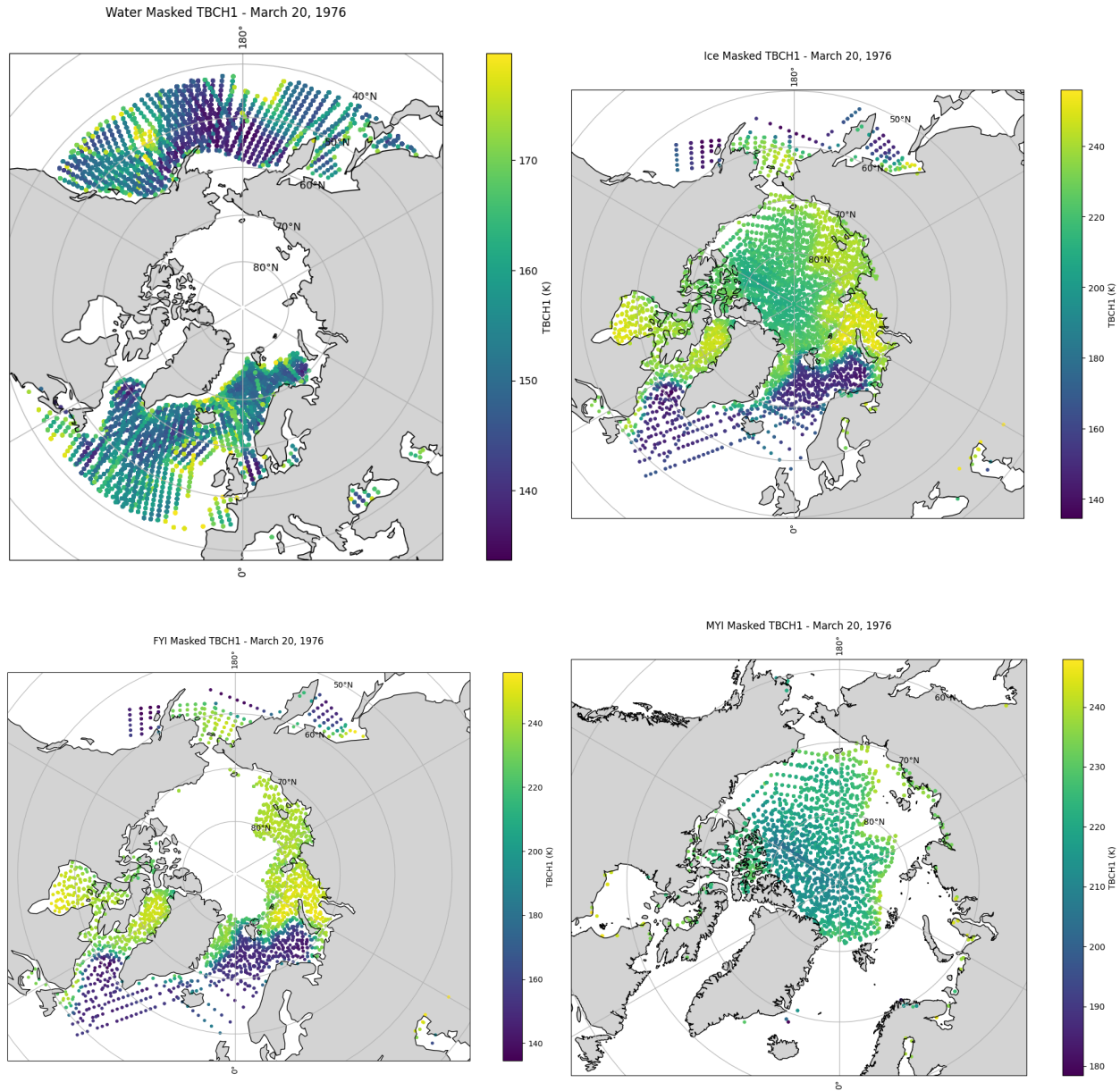


Figure 2: Water, ice, FYI, and MYI masking results (clockwise from top left). We observe successful exclusion of land areas, as well as a logical-seeming distribution of FYI vs MYI and water vs ice.

Once the raw tiepoints were obtained, they were then smoothed with a two week window to reduce the influence of anomalous daily variations, identically to the procedure described in (Kolbe, Tonboe, and Stroeve (2025)). An example result from the tiepoint smoothing is shown in figure 3. These tiepoints could then be fed into the one

channel and two channel algorithms in order to transform each measured brightness temperature into a corresponding sea ice concentration. A scatterplot of the raw smoothed tiepoints produced for the two channel algorithm is shown in figure 4.

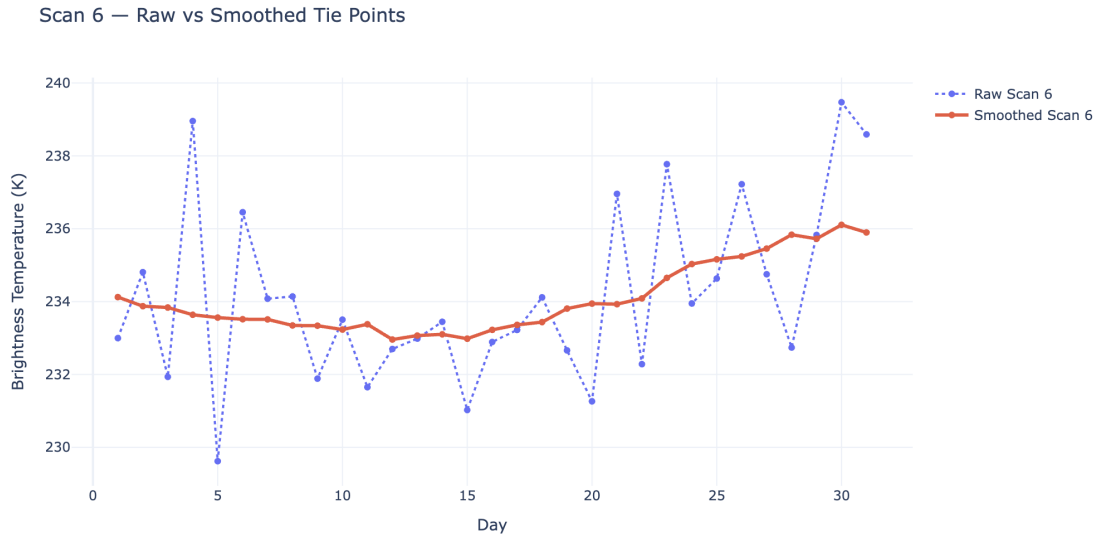


Figure 3: Example of tiepoint smoothing for the nadir footprint of the arctic ice (1 channel) tiepoint sequence for March 1976.

2.2.3 1 Channel & 2 Channel Algorithms

We find it useful to calculate sea ice concentration using both one channel and two channel algorithms, as each offers a different set of advantages. By using a one channel algorithm based on the channel 1 brightness temperatures at 22GHz, we are able to take advantage of reduced sensitivity to atmospheric noise at lower measurement frequencies (demonstrated in the sensitivity analysis in section 3.1). However, a one channel algorithm is more likely to underestimate sea ice concentration since it is incapable of differentiating between first year and multi year ice. Conversely, the two channel algorithm is vulnerable to the higher sensitivity to atmospheric noise introduced by the higher frequency channel 2 measurement at 31GHz, but is better able to distinguish between the signatures of open water and multi year ice.

The one channel algorithm is identical to the one described in (Kolbe, Tonboe, and Stroeve (2025)) is reproduced here as follows:

$$SIC_{1ch} = \frac{T_{B,obs} - T_{p,water}}{T_{p,ice} - T_{p,water}} \quad (2)$$

The two channel algorithm is adapted from the frequency mode bootstrap algorithm described in (Comiso et al. (1997)). Once preliminary sea ice concentrations were calculated using these algorithms, we then used the calculated one channel sea ice concentration as an input to the radiative transfer model described in the following section. Using

this radiative transfer model, we were able to calculate corrected brightness temperatures which were in turn fed back into the sea ice concentration algorithms to produce the corrected sea ice concentration maps shown in the section 3.3.

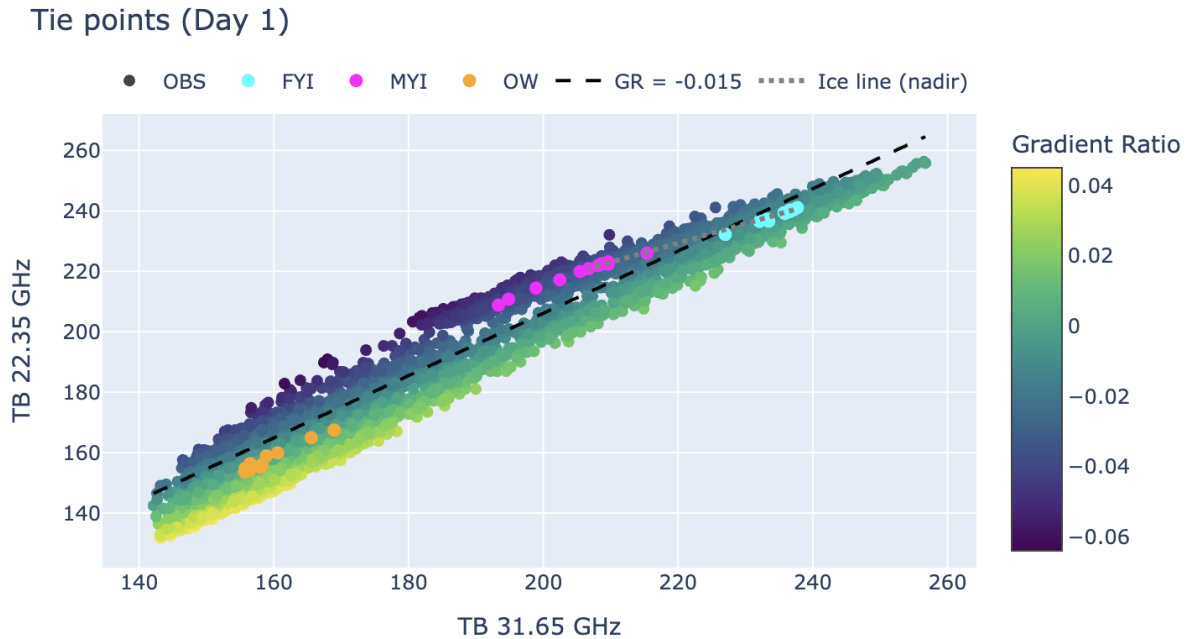


Figure 4: Scatter plot illustrating the two channel tiepoints and gradient ratio criterion for distinguishing between FYI and MYI. In this image, all observed brightness temperatures in 22.35 GHz (CH1) and 31.65 GHz (CH2) filtered by the ocean and arctic masks are plotted for 1 March 1976. There are 13 tiepoints each for ocean, FYI, and MYI corresponding to the 13 scan angles. This plot is analogous to figure 4 in Kolbe, Tonboe, and Stroeve (2025).

2.3 Atmospheric Correction via RTM

The Radiative Transfer Model RTM is used to study the radiative transfer in the atmosphere and the interaction between radiation, clouds or aerosol. In this paper we use the RTM to provide the relationship between measured brightness temperatures and geophysical parameters. A focus will be set on the sensitivities of the brightness temperature to the different parameters. The output are the vertically and horizontally polarized brightness temperatures T_V , T_H .

2.3.1 Radiative Transfer Model (RTM) Overview

This subsection is to give an overview about the theory behind the used model. The model was found by (Wentz and Meissner 2000) and uses the following input pa-

rameters: sea surface temperature T_s , wind speed W , vertically integrated water vapor V , cloud liquid water L , incidence angle θ , and sea ice concentration c_{ice} . The final AMSR brightness temperature is given by:

$$T_B(\nu) = T_V(\nu) \sin^2(\theta) + T_H(\nu) \cos^2(\theta) \quad (3)$$

Brightness Temperature Components

The vertical channel T_V (horizontal analogous) implemented in our model follows the SCAMS-adapted version of the (Wentz and Meissner 2000). The model formula used in this work is:

$$\begin{aligned} T_V(\nu) = & T_{BU}(\nu) + \tau(\nu) \left[(1 - c_{ice}) \varepsilon_V(\nu) T_s + c_{ice} e_{ice,V}(\nu) T_{I,amsr,V}(\nu) \right. \\ & + (1 - c_{ice}) (1 - \varepsilon_V(\nu)) (T_{B\Omega V}(\nu) + \tau(\nu) T_C) \\ & \left. + c_{ice} (1 - e_{ice,V}(\nu)) (T_{BD}(\nu) + \tau(\nu) T_C) \right]. \end{aligned} \quad (4)$$

, where T_{BU} is the upwelling atmospheric brightness temperature, τ the total transmittance, ε_V the ocean surface emissivity (vertical polarization), $e_{ice,V}$ the sea ice emissivity (vertical polarization), $T_{I,amsr,V}$ the effective ice temperature used by the model, $T_{B\Omega V}$ the atmospheric downwelling brightness reflected at the ocean surface, T_{BD} the atmospheric downwelling brightness reflected at sea ice and T_C the cosmic background radiation.

The RTM represents the atmosphere by five vertically integrated parameters:

1. Upwelling effective temperature T_U
2. Downwelling effective temperature T_D
3. Vertically integrated water vapor absorption A_V
4. Vertically integrated cloud water absorption A_L
5. Vertically integrated oxygen absorption A_O

The above formulation is consistent with the SCAMS geometry and is used for the sensitivity analysis in this work. The parameters are included in the definition of the different sub-terms, which describe the atmospheric influence T_{BU} , the surface emission from open water $(1 - c_{ice}) \varepsilon_V(\nu) T_s$, the emissions from ice $c_{ice} e_{ice,V}(\nu) T_{I,amsr,V}(\nu)$, the reflected sky radiation over water $((1 - c_{ice}) (1 - \varepsilon_V(\nu)) (T_{B\Omega V}(\nu) + \tau(\nu) T_C))$ and the reflected sky radiation over ice $c_{ice} (1 - e_{ice,V}(\nu)) (T_{BD}(\nu) + \tau(\nu) T_C)$. All surface and reflection terms are attenuated by the τ . The total transmittance describes all light passing through a sample combining the linear transmitted and diffuse transmitted light.

Surface Emission from Open Water The surface emissivity is controlled by the dielectric constant of seawater, which is defined in (Wentz and Meissner 2000). Basically three effects are responsible for the variation in emissivity of the surface:

1. Long-wave surface waves, which are longer than the wavelength of the radiation, mix horizontal and vertical polarization. That changes the local incidence.
2. Sea foam is a mixture of air and water that increases emissivity for both polarizations.
3. Diffraction of microwaves by small surface waves compared to the wavelength of the radiation.

For modeling, the surface can be partitioned into foam-free areas and foam-covered areas within the radiometer footprint.

Emission from Sea Ice Sea ice contributes to the observed brightness temperature through its emissivities $e_{ice,V}(\nu)$ and $e_{ice,H}(\nu)$ and its physical temperature $T_{I,amsr}$. The surface emission is weighted by the sea ice concentration c_{ice} , which partitions the footprint into open-water and ice-covered areas. Thus, the ice emission term enters the RTM analogously to the open-water contribution but scaled by c_{ice} and using the corresponding ice emissivity and temperature.

Reflected Sky Radiation Part of the atmospheric downwelling radiation is reflected at the surface. Over open water, this contribution is given by

$$T_{B\Omega}(\nu) = \left[(1 + \Omega) (1 - \tau(\nu)) (T_D - T_C) + T_C \right] R(\nu), \quad (5)$$

where $R(\nu)$ is the surface reflectivity, Ω the roughness scattering factor, T_D the downwelling atmospheric temperature, and T_C the cosmic background (2.7 K). This term appears as $T_{B\Omega V}(\nu)$ and $T_{B\Omega H}(\nu)$ in the RTM.

Over sea ice, the same expression applies but with the ice reflectivity $R_{ice}(\nu)$, giving rise to the component $T_{BD}(\nu)$ in the SCAMS implementation.

RTM Summary Concluding, the RTM combines the atmospheric emission (T_U , T_D), the atmospheric attenuation (A_O , A_V , A_L , and τ), the surface emission from water and ice and the reflected sky radiation from water and ice. The final measure is 3 and by changing the input geophysical parameters the sensitivity to each parameter can be observed.

2.3.2 Implementing the RTM into the project

The SCAMS data parameters used as inputs to the RTM are detailed in table 3.

The code for implementation of the radiative transfer model can be found on the project [github](#) in the file `seaicemapping.ipynb`. The correction is applied as follows:

$$T_{B,corr} = T_{B,meas} + (T_{B,ref} - T_{B,calc}) \quad (6)$$

Where $T_{B,meas}$ is the SCAMS measured brightness temperature, and $T_{B,calc}$ is the value calculated by running the RTM value on the corresponding pixel for $T_{B,meas}$. $T_{B,ref}$ is a reference value produced by running the RTM model on the mean values for V, W, L, T_a , and T_s for a the given region (arctic vs antarctic) and a constant incidence angle of 55° . Figure 10 shows the difference between the 2 channel sea ice concentration values before and after RTM correction.

2.4 Resampling and Masking

The raw data from the SCAMS instrument are datapoints in swaths corresponding to the area covered by the 13 scan angles from the instrument. In order to analyze the sea ice concentration and sea ice types it is necessary to resample the data on to a fixed grid. The python package "pyresample" was used to convert the swath data to the EASE 1 grid. A land mask and a polar mask, excluding everything above -40 degrees south and below 40 north, were applied for the sea ice concentration and ice type classification before resampling. The code for both resampling and masking can be found on [github](#) in the file `seaicemapping.ipynb`.

3 Results

3.1 Sensitivity Analysis

To conduct the sensitivity analysis, the input parameter of the RTM were changed individually, while the others were kept constant at a polar reference state. Objective is, to see the change in the brightness temperature. The results are plotted in figure 5. 5a shows the sensitivity of T_B to water vapor V , which is moderate, reflecting absorption near the 22.235 GHz vapor line, where the linear sensitivity is higher than for the 31 GHz channel. The sensitivity to water vapor is particularly in the observed dry polar region much weaker than the sensitivity to cloud liquid water 5c, which produces the strongest change in T_B among all parameters. Globally observed, the sensitivity to water vapor is expected to be higher than the sensitivity to cloud liquid water. This behavior is consistent with the dominant absorption of cloud droplets at microwave frequencies. Sensitivity is substantially higher at 31 GHz than at 22 GHz. The sensitivity to the other four parameters is similar for both channels: Increasing wind speed W (5b) causes ocean surface roughness and foam coverage, which results in reduced reflectivity and an increasing emissivity. That again raises T_B (Wentz 1997). The 22 GHz channel shows slightly stronger surface sensitivity than 31 GHz, but they are almost parallel. Figure 5f shows the increase of the brightness temperature with the incidence angle. That is due to the fact that the Fresnel reflectivity decreases and emissivity increases at larger viewing angles (Meissner and Wentz 2012). The sea surface temperature T_s (5e) shows weak sensitivity near the freezing range 271 K to 276 K, because the seawater dielectric constant varies only mildly with temperature in this region (Meissner and Wentz 2012).

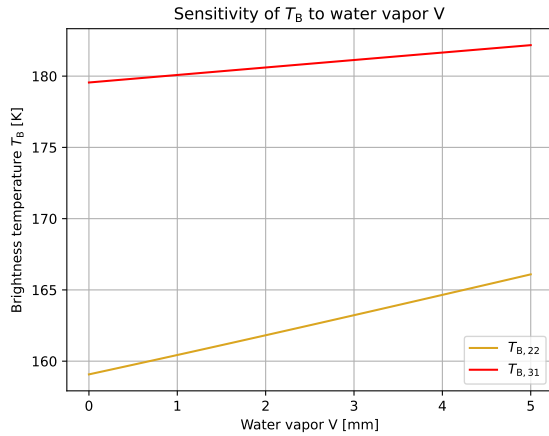
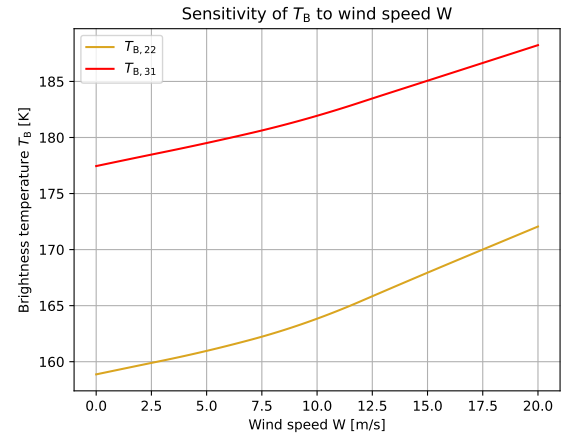
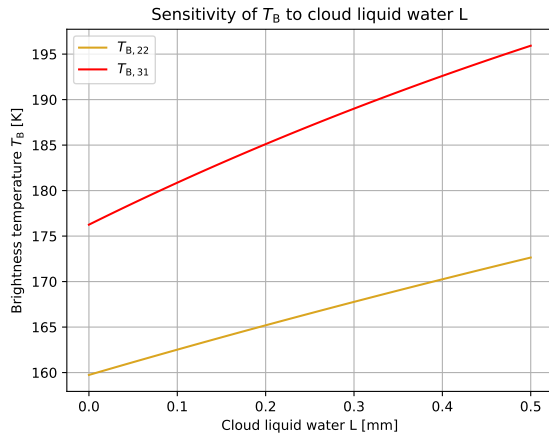
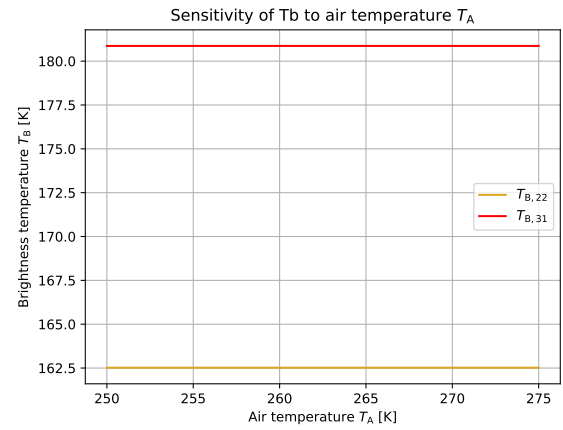
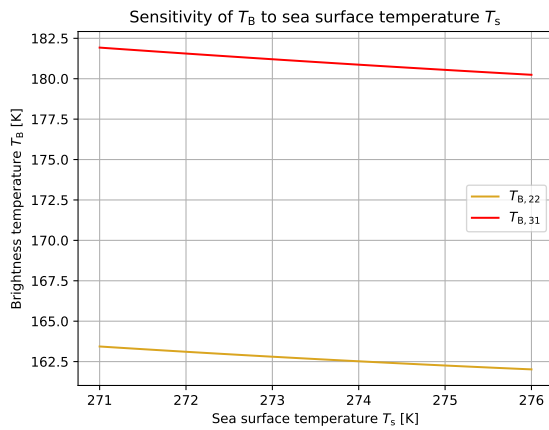
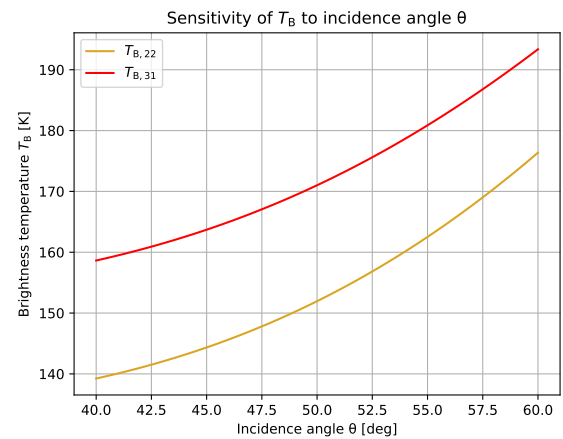
(a) Water vapor V [mm](b) Wind speed W [m/s](c) Cloud liquid water L [mm](d) Air temperature T_a [K](e) Sea surface temperature T_s [K](f) Incidence angle θ [deg]

Figure 5: Sensitivity of simulated brightness temperatures T_B at 22 and 31 GHz to key geophysical parameters in polar conditions: (a) water vapor V , (b) wind speed W , (c) cloud liquid water L , (d) air temperature T_a , (e) sea surface temperature T_s , and (f) incidence angle θ .

The air temperature T_a , plotted in figure 5d, has the weakest influence on T_B in cold and dry polar atmospheres because atmospheric emission is small (Rosenkranz 1998). Variations in T_a mainly affect the upwelling atmospheric term, but only marginally at these frequencies (Janssen 1993).

Figure 6 shows the ranking of the absolute brightness temperatures changes per unit. It is clearly visible, that cloud liquid water produces the strongest absolute variations in brightness temperature. On a global scale, it is expected that water vapor would have a higher absolute influence. However, due to the dry and cold temperatures observed in the polar regions, the effect is lower than that of cloud liquid water.

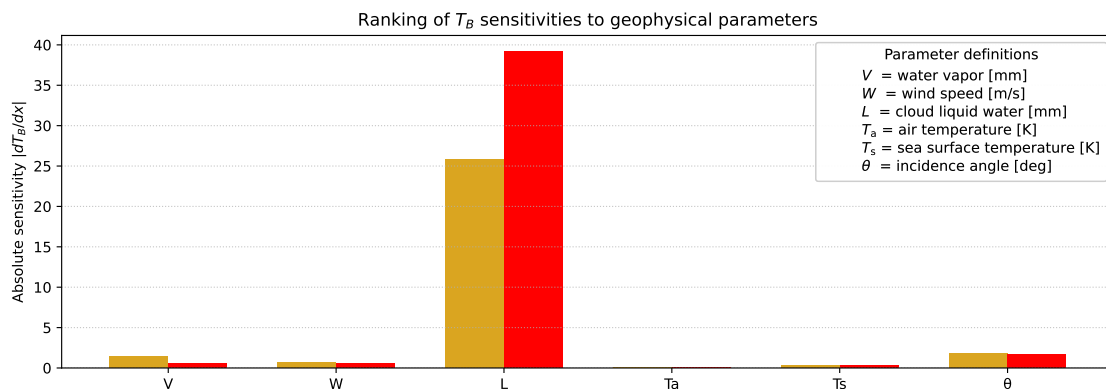
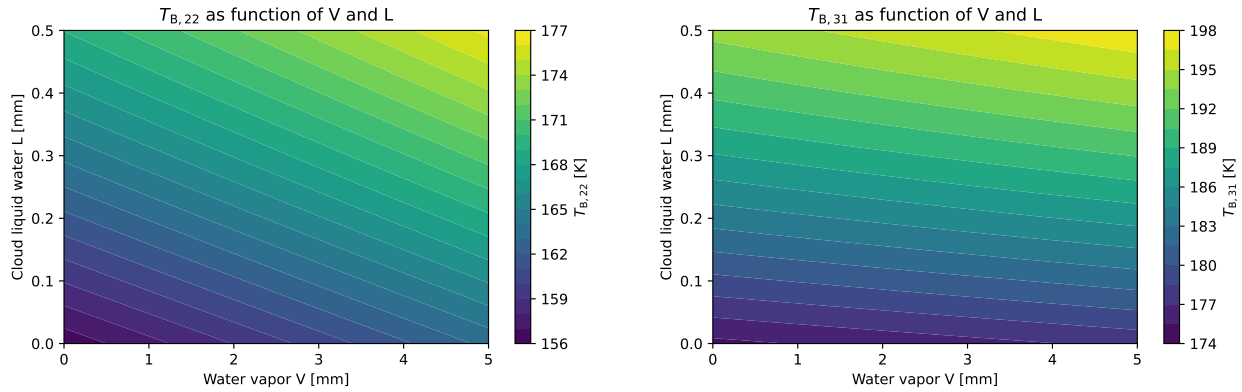


Figure 6: Ranking of local linear sensitivities $|dT_B/dx|$ for the 22 GHz and 31 GHz channels, evaluated around a polar reference state. The legend summarizes the definitions of all geophysical parameters.

Based on the one-dimensional sensitivity analysis, it becomes clear that each geophysical parameter affects the brightness temperature in a characteristic way, with cloud liquid water, incidence angle, and wind-related surface effects dominating the variability. However, in realistic atmospheric and oceanic conditions, these parameters do not vary independently. Their combined influence can introduce nonlinear interactions and modify the apparent sensitivity of individual terms. To investigate such effects, we decided to extend the sensitivity analysis and evaluate the brightness temperature as function of two geophysical parameters simultaneously. That allows us to assess how parameter interaction influence the sensitivity of the brightness temperature.

Figure 7 shows the combined influence of cloud liquid water and water vapor for both the 22 GHz and the 31 GHz channel. As observed before, cloud liquid water produces the strongest variations in brightness temperature, as indicated by steep gradients in the L direction. Water vapor mainly shifts the overall baseline of T_B , resulting in diagonal structures linked to the combined effects of absorption and surface emission. The 22 GHz channel shows a stronger response to both parameters than the 31 GHz channel, which is due to the 22 GHz channel being close to the water absorption line at 22.235 GHz. Even for low values typical of polar environments, the atmospheric contribution remains clearly visible.

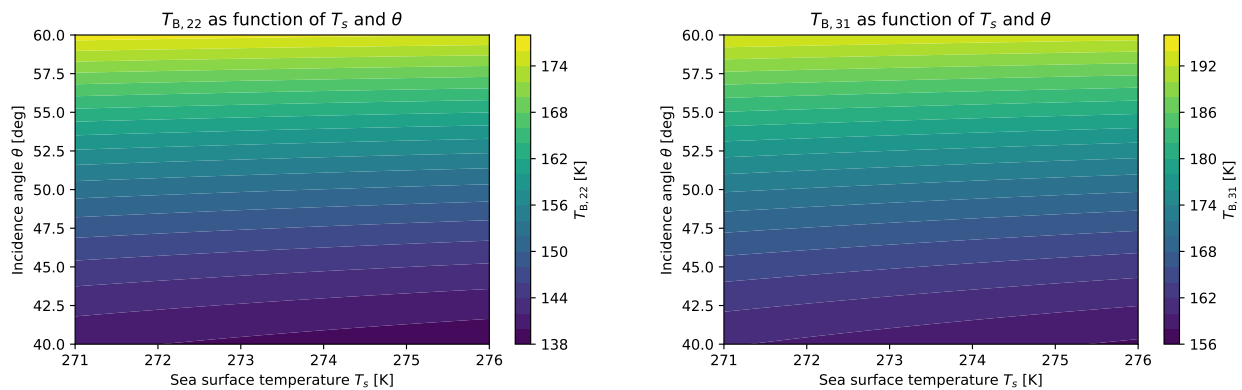


(a) $T_{B,22}$ as function of water vapor V and cloud liquid water L .

(b) $T_{B,31}$ as function of water vapor V and cloud liquid water L .

Figure 7: Brightness temperature sensitivity to the combined effects of water vapor V and cloud liquid water L for the 22 GHz and 31 GHz channels.

Incidence angle θ is the dominant factor in the combined response, with increasing θ enhancing surface emissivity and thereby raising T_B . Sea surface temperature near the freezing point causes only moderate changes. Figure 8 The 22 GHz channel exhibits slightly stronger surface sensitivity than the 31 GHz channel. The smooth contour structure suggests nearly linear variation with both parameters, and the dependence on θ underscores the importance of accurate viewing geometry in microwave retrievals.



(a) $T_{B,22}$ as function of sea surface temperature T_s and incidence angle θ .

(b) $T_{B,31}$ as function of sea surface temperature T_s and incidence angle θ .

Figure 8: Brightness temperature sensitivity to the combined effects of sea surface temperature T_s and incidence angle θ for the 22 GHz and 31 GHz channels.

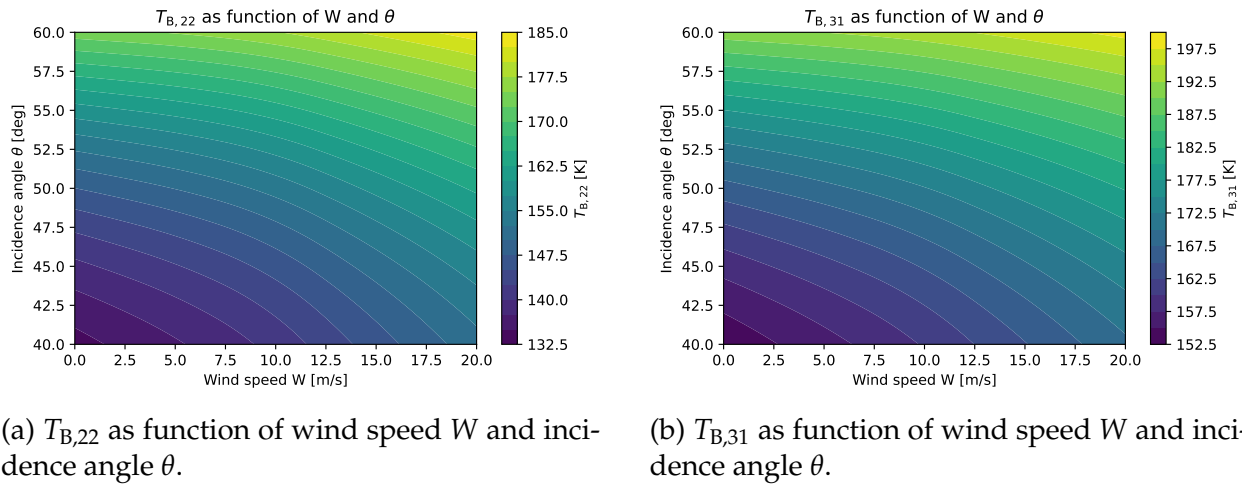


Figure 9: Brightness temperature sensitivity to the combined effects of wind speed W and incidence angle θ for the 22 GHz and 31 GHz channels.

Both wind speed W and incidence angle θ increase ocean surface emissivity, leading to higher T_B . Sensitivity to wind speed is more pronounced at 22 GHz, which responds more strongly to roughness and foam coverage than the 31 GHz channel as indicated in figure 9. The variation with θ remains significant for both channels, reflecting the geometric influence on emissivity. Higher wind speeds reduce surface reflectivity through increased roughness and foam generation. Together, the combined effects of W and θ explain much of the surface-driven variability observed in microwave brightness temperatures over open water.

3.2 Noise Reduction Evaluation

The difference between the corrected and uncorrected sea ice concentrations for the 1 and 2 channel algorithms is plotted in figure 10. Here, we observe that the correction from the RTM generally results in an increase in the SIC, with the magnitude of the correction noticeably larger in the ocean regions than in the icecaps.

Figure 11 shows the standard deviation of the corrected vs uncorrected brightness temperatures as a function of the incidence angle for a chosen sample region of arctic water for 01 March 1976. The key observation here is that for both channel frequencies, the standard deviations are reduced by the atmospheric correction, which is a validation that it is producing the desired effect. We also see that the standard deviations are roughly symmetrical about the nadir, which is to be expected since the nadir suffers from the smallest amount of atmospheric interference, which increases as a function of incidence angle. We also observe a decrease in standard deviation at the most extreme incidence angles, likely because the footprint size becomes so large that it effectively reduces the standard deviation by creating a larger sample region. We would expect the correction to have a more modest effect for the higher frequency since the lower frequency is less sensitive to atmospheric water vapor, but this does not necessarily appear to be the case,

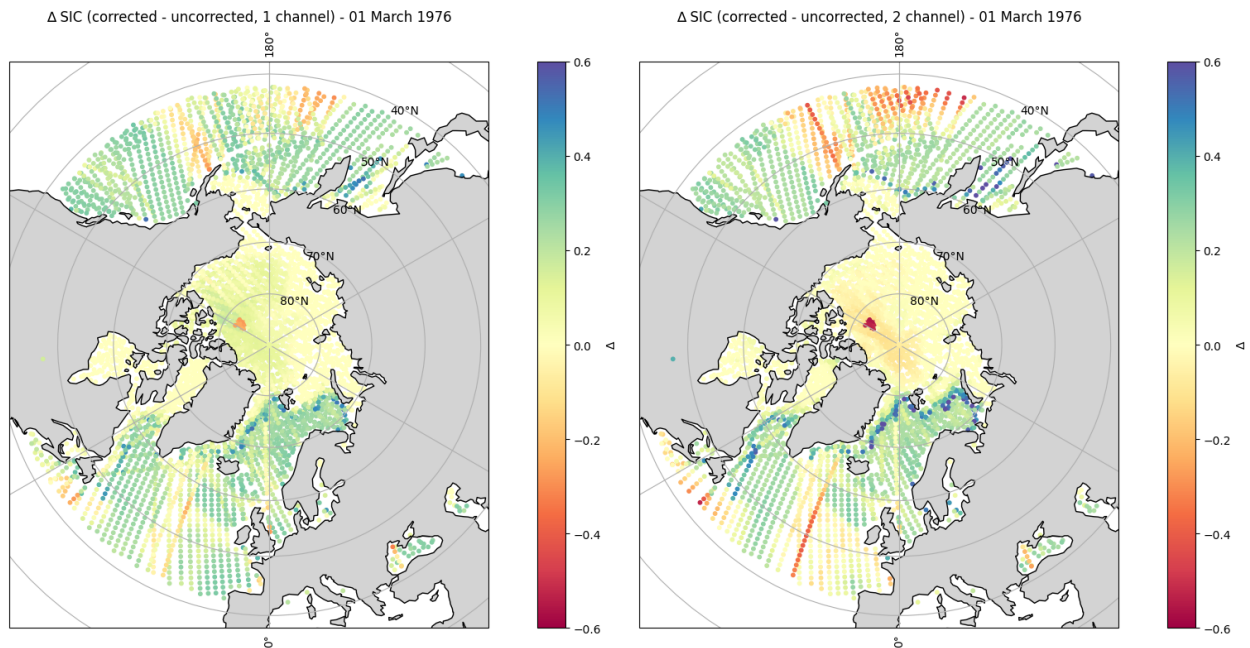


Figure 10: Difference map for the corrected vs uncorrected 1 and 2 channel sea ice concentrations for one day in the sample period (01 March 1976).

at least for this sample. We also observe that the standard deviations are much higher than those in Kolbe, Tonboe, and Stroeve (2025). We hypothesize that this could be because our geographic sample region is rather large, spanning from 55 to 60 ° latitude.

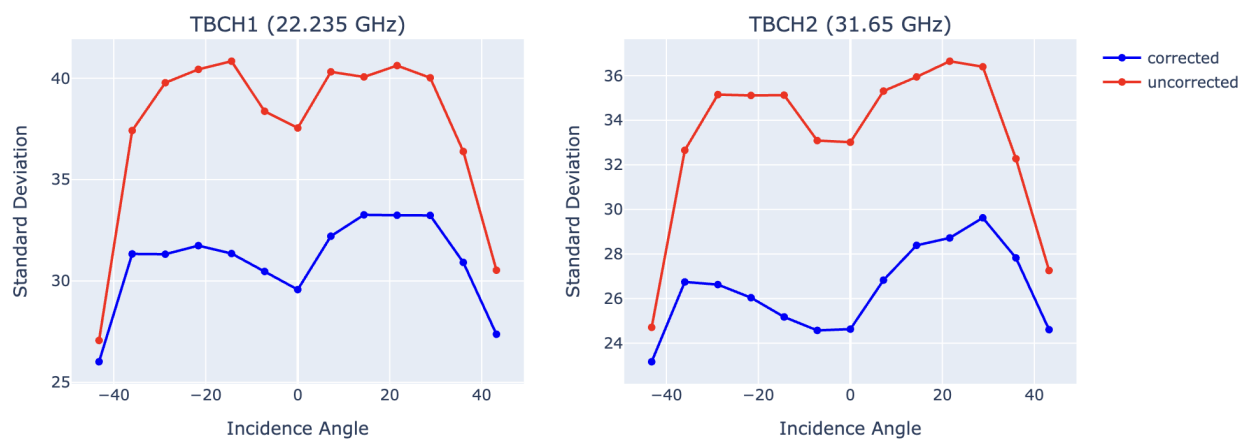


Figure 11: Illustration of standard deviation of corrected vs uncorrected brightness temperatures in arctic water in the sample region from 55 to 80 ° latitude for 01 March 1976. Analogous to figure 5 in Kolbe, Tonboe, and Stroeve (2025)

3.3 Sea Ice Concentration Maps

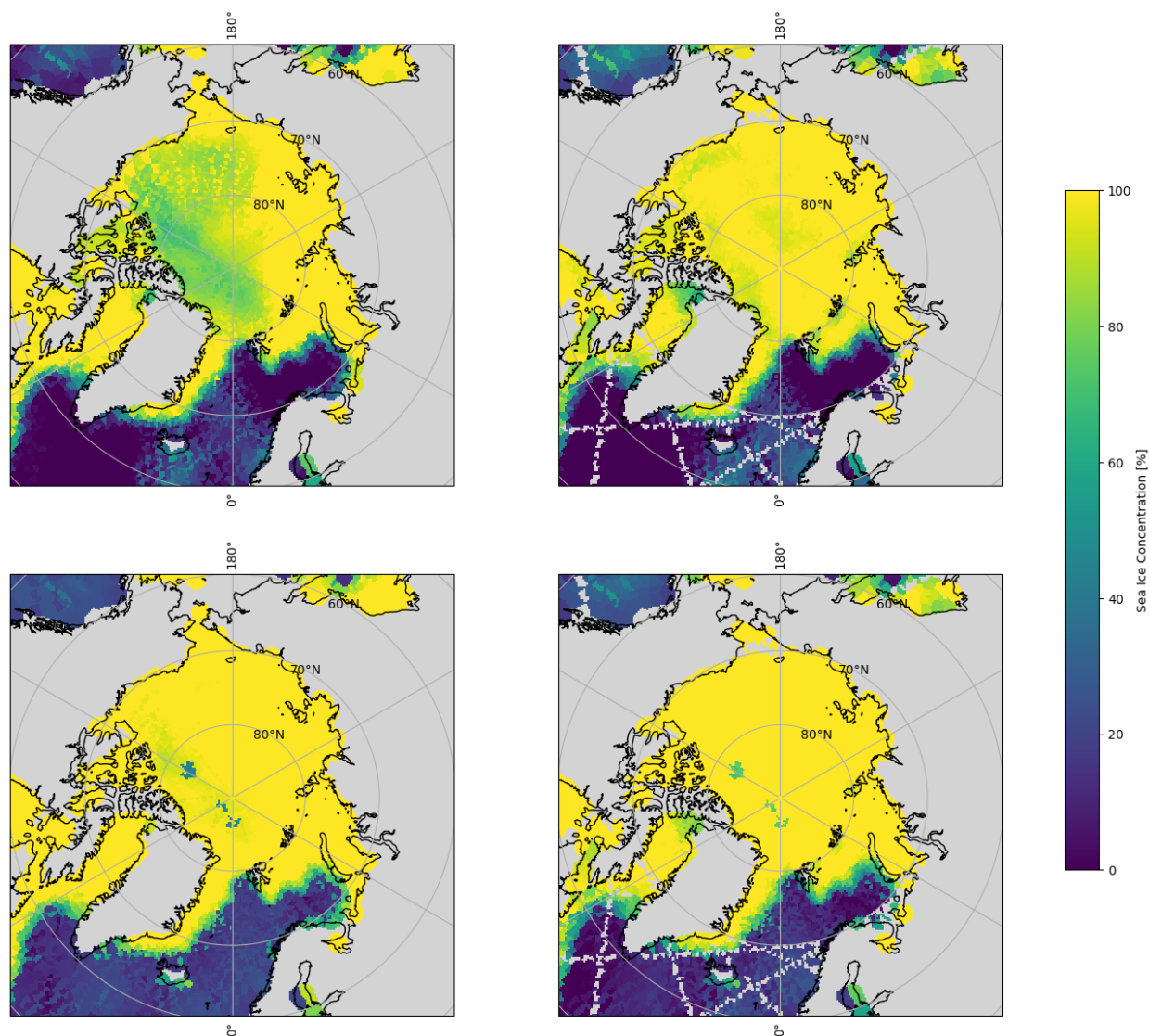


Figure 12: Map over SIC in the Arctic, March 1st 1976. Top left: Uncorrected SIC calculated with the one-channel algorithm. Top right: Uncorrected SIC calculated with the two-channel algorithm. Bottom left: Corrected SIC calculated with the one-channel algorithm. Bottom right: Corrected SIC calculated with the two-channel algorithm.

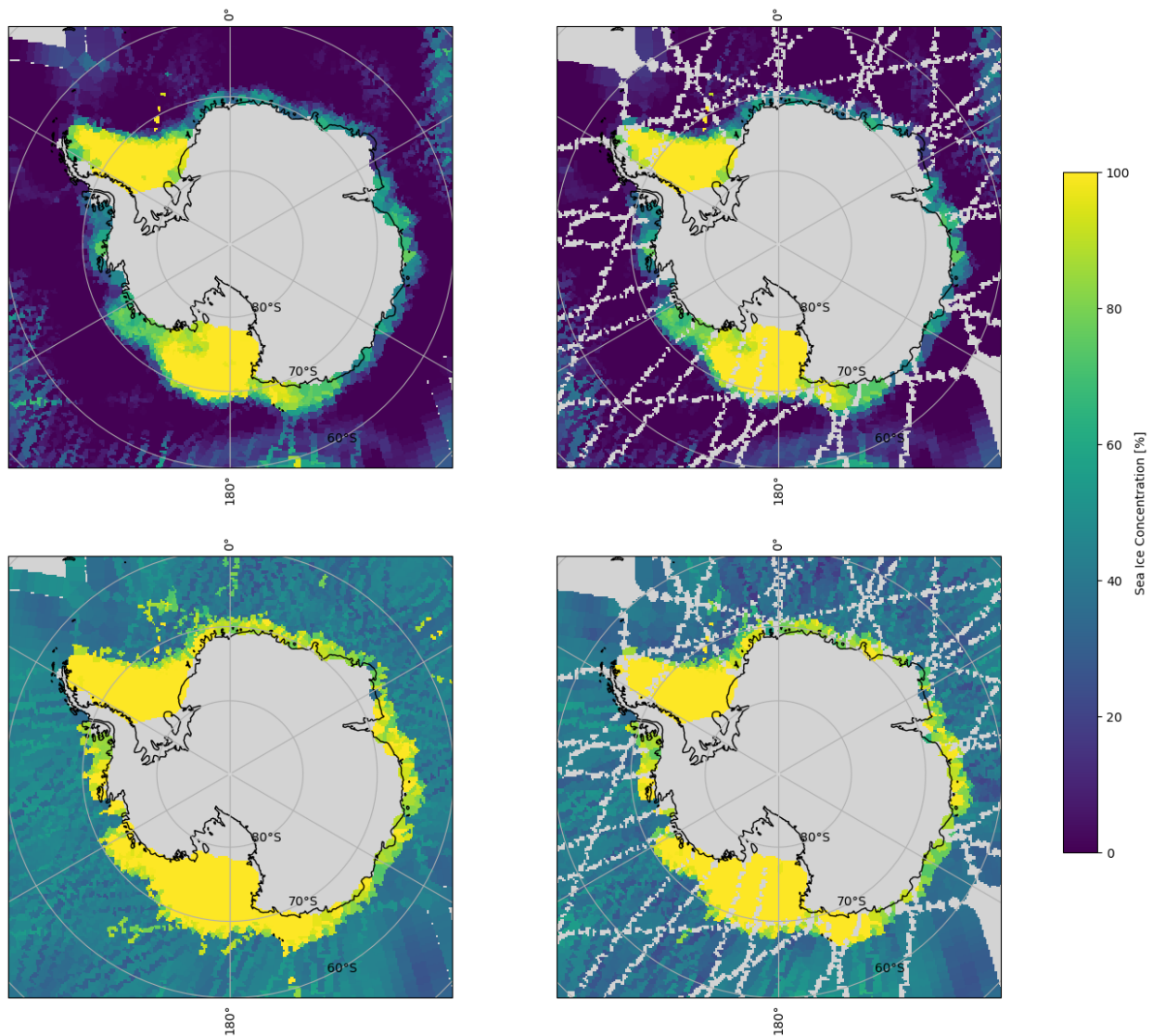


Figure 13: Map over SIC in the Antarctic, March 1st 1976. Top left: Uncorrected SIC calculated with the one-channel algorithm. Top right: Uncorrected SIC calculated with the two-channel algorithm. Bottom left: Corrected SIC calculated with the one-channel algorithm. Bottom right: Corrected SIC calculated with the two-channel algorithm.

3.4 Sea Ice Concentration Results

Figures 12 and 13 illustrate the sea ice concentration in the northern and southern hemisphere for the 1st of march 1976. There is a notable difference between the one-channel and two-channel algorithm plots. There is a clear underestimation of SIC in the top left plot in Figure 12. The one-channel algorithm uses SCAMS data and tie points from the 22 GHz channel, and this algorithm is therefore not able to take the differences in brightness temperature coming from the different ice types into account when calculating

the SIC. This aligns well with the results from Kolbe, Tonboe, and Stroeve (2025). This is further supported by the top right plot, which illustrates the uncorrected SIC calculated with the two-channel algorithm. It is clear that the two-channel algorithm can categorize MYI as 100% sea ice.

In the corrected SIC plots (the two bottom plots) of the northern hemisphere we observe a small area of what looks like $SIC < 40\%$ in an area where one would expect 100% SIC. This feature could represent an Arctic polynya, a natural phenomenon in which areas of open water form within the surrounding sea ice (Mooers and Smith (1980)). However, it is also plausible that the pattern arises from some limitations in the data or imperfections in the RTM. Furthermore, the highest correction in both the one-channel and two-channel plots is observed over open water.

The maps display a daily SIC calculation and the short sampling period is likely the reason why clear stripes appear in all plots using the two-channel algorithm. A potential approach to resolving this issue could be to look at a weekly average instead of a daily result.

3.5 Sea Ice Type Results

The ice types are determined by the gradient threshold defined in equation 1.

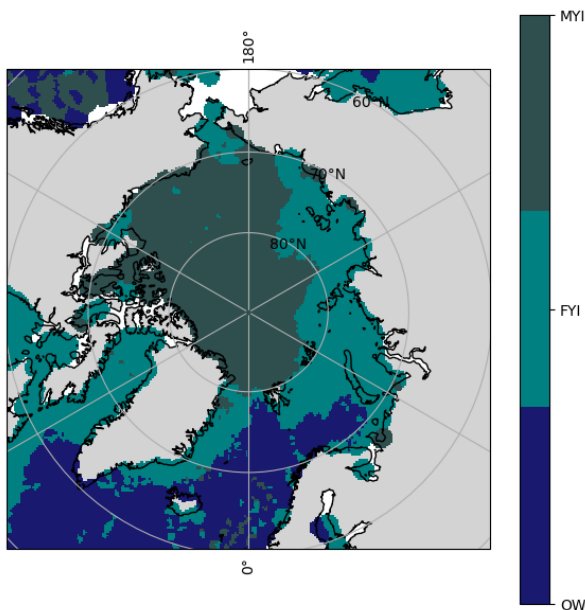


Figure 14: Map over first year ice (FYI) and multi year ice (MYI) in the northern hemisphere, March 1st 1976.

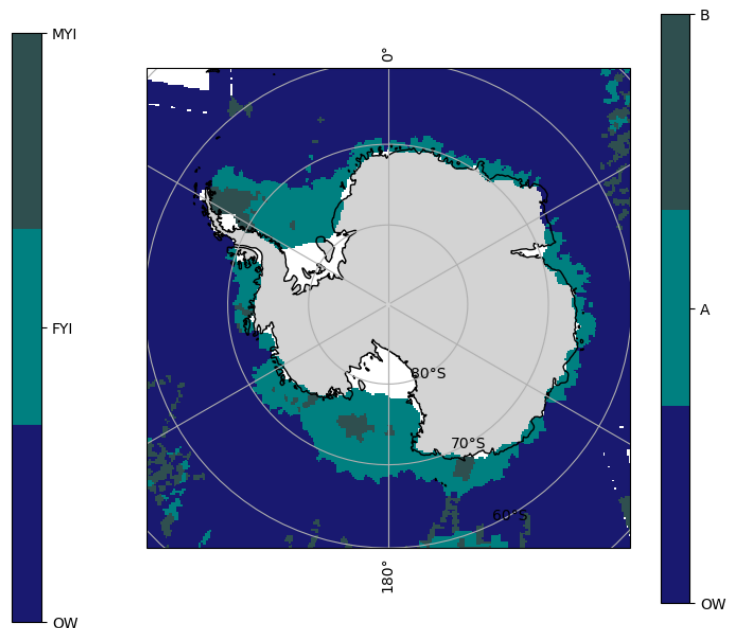


Figure 15: The sea ice in the southern hemisphere is typically denoted A (FYI) and B (MYI), here for March 1st 1967. (Kolbe, Tonboe, and Stroeve (2025)).

A footprint with first year ice has a gradient over -0.015 , and a footprint with multi year ice a gradient under -0.015 . The different ice types on March 1st 1976 are displayed

in figure 14 and 15 for the northern and southern hemisphere. 14 is a result of the classification of ice types using the gradient ratio criterion, FYI and MYI tie points illustrated in figure 4. The map also includes a mask where areas with less than 30% sea ice concentration is set to open water (OW). The figure includes an ocean mask where footprints with $SIC > 30\%$ is classified as open water. We observe some small regions in both figures that appear to be areas of MYI, although this is likely not correct.

4 Discussion

The sensitivity analysis shows that cloud liquid water has the strongest impact on brightness temperature in the polar region at 22 GHz and 31 GHz, with an influence that clearly exceeds that of water vapor as well as that of wind speed and temperature-related parameters. Therefore, accurate correction for cloud liquid water, but also the other parameters is essential to avoid misinterpreting atmospheric effects as for example changes in ice concentration. The comparatively weak sensitivity to water vapor indicates that it plays a minor role in retrieval uncertainty under polar conditions. Meanwhile, the channel-dependent sensitivities highlight the importance of carefully selecting and combining frequencies to distinguish between open water and sea ice.

Applying the RTM correction to the SCAMS observations substantially reduces the standard deviation of brightness temperatures across incidence angles, demonstrating that the correction effectively removes part of the atmospheric and surface-induced variability. The effect is strongest over open water, where atmospheric contributions are largest. However, TB variability remains higher than that observed in (Kolbe, Tonboe, and Stroeve (2025)), which could be due to the large spatial region used in the evaluation.

The sea ice concentration maps show that the two-channel algorithm performs better than the one-channel algorithm in regions containing multi-year ice, reducing the underestimation seen at 22 GHz. The RTM correction increases SIC primarily over open water, where uncorrected brightness temperatures are biased low. A localized low-SIC feature in the central Arctic could represent a real polynya or numerical errors in the atmospheric correction.

One notable deviation we see from (Kolbe, Tonboe, and Stroeve (2025)) is that the underestimation of sea ice in the one-channel algorithm appears to be mostly resolved by the RTM, which we would not necessarily expect since the RTM is also only using the 1 channel sea ice concentration data. One reason for this could be that the RTM is actually *overcorrecting*: we observe that the RTM correction generally results in an increase in brightness temperature, which makes sense considering that water vapor in the atmosphere artificially lowers the surface brightness temperature we are seeking to measure. If the magnitude of this correction is greater than that of the actual influence of the water vapor, it would make sense that we then observe a much higher prevalence of 100% SIC in the 1 channel algorithm than we would otherwise expect.

Finally, the ice-type classification largely captures the expected FYI/MYI patterns but shows scattered misclassifications near the ice edge, likely due to noise or threshold sensitivity. The maps could be further improved by including data from a longer sampling

period to increase geographical coverage.

Overall, the results indicate that the RTM correction improves physical consistency and reduces noise in the TB fields. There are several areas where refinement of the processing pipeline could yield further improved results, from adjustment of the RTM reference value parameters to more precise tiepoint selection.

5 Conclusion

Continuous monitoring of sea ice in the Arctic and Antarctic is essential for understanding how climate change impacts the earth's system. (World Meteorological Organization (2011)). Using the noise reducing radiative transfer model together with the historical NIMBUS 6 satellite SCAMS data, and following the methodology described above, the group successfully calculated the sea ice concentrations and sea ice types over both polar regions in March 1976.

The sensitivity analysis showed that cloud liquid water was the most dominant atmospheric parameter. Its influence clearly exceeded the influence of the other parameters. These findings are crucial for interpreting the results as variations cloud liquid water can cause similar variations in the brightness temperature as those between open water and ice. Applying the RTM reduced the standard deviation of the brightness temperature, which shows that removing a big part of the atmospheric and surface-related variability was successful. The differences between the one-channel and two-channel algorithm were as expected. The one-channel algorithm proved less sensitive to atmospheric noise, and the two-channel algorithm allowed for distinguishing between different ice types. Future work could include extending the data record to weekly or monthly averages, to further reduce noise and daily variability from the plots. Another future improvement could be a comparison to the manually produced National Ice Center (NIC) weekly ice charts as done in (Kolbe, Tonboe, and Stroeve (2025)).

6 Authorship & AI Acknowledgement

This project was completed collaboratively by all group participants. The primary responsibility for authorship of this paper was divided as shown in table 1.

Group Member	Sections Authored
Jenny Brun	2, 2.1, 2.4, 3.3, 3.4, 3.5, 5
Dahlia Dry	Abstract, 2, 2.2, 3.2, 4, Appendix A
Hannah Gerlach	1, 2.3, 3.1, 4, 5

Table 1: Distribution of project responsibility among group mates.

AI tools were used to some extent in the image analysis process and in the writing of this paper. The specific ways in which AI was used by each group mate is detailed as follows:

Jenny Brun: ChatGPT was used to support the development of some Python plotting scripts and to format bibliographic sources into LaTeX/BibTeX entries.

Dahlia Dry: I used GitHub Copilot to assist in writing the C++ code involved in the image analysis. I used ChatGPT to generate BibTeX references for the sources cited. However, the writing in this report is entirely my own.

Hannah Gerlach: In the preparation of this report, I used two different AI-based tools to improve efficiency and clarity. ChatGPT was used to convert my handwritten equations into correctly formatted LaTeX code, to create tables and figure templates suitable for Overleaf. Additionally, I used DeepL to review and correct the grammar and spelling of my text. These tools supported the technical writing process, but all analysis, interpretation, and conclusions presented in this report are entirely my own.

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A Supplementary Tables

Variable	Dataset Name	Units	Use in Project
Brightness Temperature (22 GHz)	TBCH1	K	Low-frequency channel used for one-channel SIC and Comiso two-channel SIC.
Brightness Temperature (31 GHz)	TBCH2	K	Higher-frequency channel used for gradient ratio (GR) and two-channel SIC.
Latitude	LAT	degrees	SCAMS beam geolocation used to mask land, determine hemisphere, and plot swath datasets.
Longitude	LON	degrees	SCAMS beam geolocation used together with latitude for masking and mapping.
Surface Elevation	ELEV (ERA5)	km	Used to identify land/ice sheets (ELEV > 0.05 km) for constructing a correct ocean mask.
Land–Sea Mask (ERA5)	lsm	0–1	ERA5 grid land mask, used as auxiliary ocean mask (but on different grid than SCAMS beams).
Sea Ice Concentration (ERA5)	siconc	0–1	Reference dataset used for comparison with retrieved SIC and for FYI/MYI tiepoint masks.

Table 2: Base SCAMS+ERA5 dataset variables used for mapping and tiepoint calculation

Variable	Dataset Name	Units	Use in Project
Column Water Vapour	H2OVAP	mm	Input to the RTM as V.
Column Liquid Water	H2OLIQ	mm	Input to the RTM as L
U Wind (10 m)	u10	m s^{-1}	Contributes to surface wind speed (W) in the RTM, which modifies surface emission and reflection.
V Wind (10 m)	v10	m s^{-1}	Used with u10 to compute surface wind speed ($W = \sqrt{u^2 + v^2}$).
2-m Air Temperature	t2m	K	Used as atmospheric temperature (T_a) input to the RTM.
Sea Surface Temperature	sst	K	Used for the T_s input to the RTM.
Ice Temperature (layer 1)	istl1	K	Used as near-surface ice temperature for RTM input (T_i).
Incidence Angle	INANG	degrees	Beam incidence angle for each of the 13 SCAMS scan positions; used in RTM.

Table 3: Supplementary ERA5 dataset variables used for inputs to radiative transfer model

Mask Type	Conditions
Ocean	<code>lsm == 0</code> <code>ELEV > 0.05</code>
Arctic	<code>42 <= LAT <= 90</code>
Antarctic	<code>-42 >= LAT >= -90</code>
Water	<code>siconc == 0</code> <code>ocean & 90 < TBCH1 < 180</code> <code>90 < TBCH2 < 180</code>
Ice (1 channel)	<code>siconc == 0</code> <code>ocean & 100 < TBCH1 < 274</code> <code>100 < TBCH2 < 274</code>
FYI (2 channel)	<code>ice & GR >= ?</code> <code>not((n13_obs == 0 n13_obs == 12) & LAT < 80)</code>
MYI (2 channel)	<code>ice & GR < ?</code> <code>not((n13_obs == 0 n13_obs == 12) & LAT < 80)</code>

Table 4: Tie-point mask definitions used in the sea-ice retrieval algorithm, where variable names from the SCAMS dataset are described in table 2 (`siconc` refers to the ERA5 sea ice concentration model).